DESIGN AND IMPLEMENTATION OF A DBMS FOR RECRUITMENT AND APPLICANT TRACKING SYSTEM

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1.PROBLEM IDENTIFICATION

- Recruitment information is often scattered across emails, spreadsheets and separate records.
- It becomes difficult to maintain a complete history of each candidate.
- Manual tracking can lead to duplicate data, missed interviews and inconsistent application statuses.
- HR departments need a centralized system to manage:
  -Vacancies
  -Candidates
  -Applications
  -Interviews
  -Evaluations
  -Offers
  -Joining records
- Therefore, we propose a relational database system that connects the complete recruitment lifecycle

2.SCOPE OF THE SYSTEM

Vacancy management

- Create and manage job vacancies
- Define required positions and jobs
- Track vacancy status

Candidate Management

•Store candidate information.
•Maintain skills and qualifications.
•Keep candidate records organized.

Application Tracking

•Record applications for vacancies.
•Track application status.
•Manage shortlisting and selection.

Interview Management

- Create interview rounds.
- Assign interview panels.
- Schedule interviews.
- Record evaluations.

Offer & Joining

- Generate offers for selected candidates.
- Track offer status.
- Record joining information.

Reporting

- Generate recruitment reports.
- Track vacancy progress.
- Analyze applications, interviews and hiring.

3.PROJECT OBJECTIVES

- **Centralize Recruitment Data**
  To store all recruitment related data under one database.
- **Reduce Data Redundancy**
  To normalize the data upto the Third Formal Form (3NF)
- **Maintain Data Integrity**
  Use Primary keys, Foreign keys, UNIQUE constraints, NOT NULL constraints and DEFAULT values
- **Track Candidate Progress**
  Follow candidates throughout the whole process(ie-application>interview>evaluation>offer>joining)
- **Support Reporting and Analysis**
  Use SQL queries, joins, aggregate functions, nested queries and views
- **Provide a Foundation for a Front End**
  Develop a small application using typescript

4.USERS OF THE SYSTEM

HR / Recruiter

- Creating and managing vacancies
- Reviewing applications
- Shortlisting candidates
- Scheduling interviews
- Recording evaluations
- Issuing offers
- Tracking joining status

Interview Panel

- View assigned interviews
- Conduct interview rounds
- Evaluate candidates
- Enter scores and remarks
- Provide recommendations

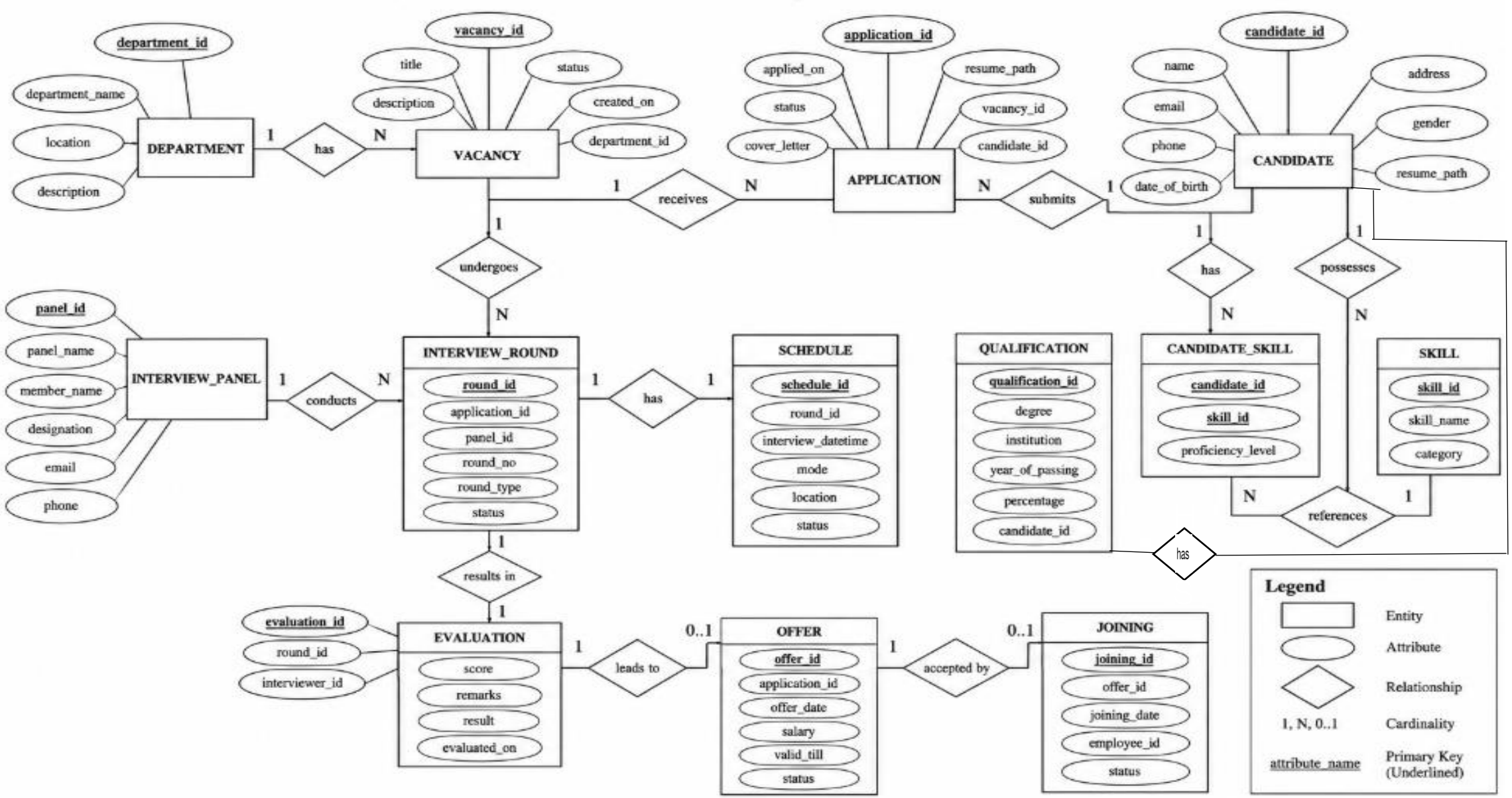
Administrator / Management

•Monitor recruitment progress
•View vacancy status
•Review offers and joining records
•Generate recruitment reports
•Maintain important system data

5.FUNCTIONAL REQUIREMENTS

- **Create and manage** job vacancies.
- **Register** candidates with their personal information.
- **Store** candidate skills and qualifications.
- **Create and track** applications.
- **Shortlist** candidates for suitable vacancies.
- **Create and manage** interview rounds.
- **Assign** interview panels.
- **Schedule** interviews with appropriate dates and times.
- **Record** evaluation scores, remarks and results.
- **Create** offers for selected candidates.
- **Record** the candidate's joining information.
- **Generate** useful reports such as:
  -Applicant reports
  -Shortlisted candidates
  -Interview schedules
  -Offer status
  -Vacancy progress

6.ER DIAGRAM- APPLICANT TRACKING SYSTEM



WEEK 1 DELIVERABLES

- Problem Identification
- Scope
- Objectives
- Users
- Functional Requirements
- Conceptual design
- ER Components and Diagram

WEEK 2 (NEXT WEEK) DELIVERABLES

- Database Normalization
- Define Constraints
- Add Business Rules
- Populate Sample Data
- Develop SQL Queries
- Develop Front End
- Testing

THANK YOU